Table 2 All models evaluated in AbstentionBench

Name	HF ID	Reasoning
DeepSeek R1 Distill Llama 70B	deepseek-ai/DeepSeek-R1-Distill-Llama-70B	✓
GPT-4o (2024-10-21)	-	✗
Gemini 1.5 Pro	-	✗
Llama 3.1 405B Instruct	meta-llama/Llama-3.1-405B-Instruct	✗
Llama 3.1 70B Base	meta-llama/Llama-3.1-70B	✗
Llama 3.1 70B Instruct	meta-llama/Llama-3.1-70B-Instruct	✗
Llama 3.1 70B Tülu 3 DPO	allenai/Llama-3.1-Tulu-3-70B-DPO	✗
Llama 3.1 70B Tülu 3 PPO RLVF	Llama-3.1-Tulu-3-70B	✓
Llama 3.1 70B Tülu 3 SFT	allenai/Llama-3.1-Tulu-3-70B-SFT	✗
Llama 3.1 8B Base	meta-llama/Llama-3.1-8B	✗
Llama 3.1 8B Instruct	meta-llama/Llama-3.1-8B-Instruct	✗
Llama 3.1 8B Tülu 3 DPO	allenai/Llama-3.1-Tulu-3-8B-DPO	✗
Llama 3.1 8B Tülu 3 PPO RLVF	Llama-3.1-Tulu-3-8B	✓
Llama 3.1 8B Tülu 3 SFT	allenai/Llama-3.1-Tulu-3-8B-SFT	✗
Llama 3.3 70B Instruct	meta-llama/Llama-3.3-70B-Instruct	✗
Mistral 7B Instruct v0.3	mistralai/Mistral-7B-Instruct-v0.3	✗
OLMo Instruct 7B	allenai/OLMo-7B-0724-Instruct-hf	✗
Qwen2.5 32B	Qwen/Qwen2.5-32B-Instruct	✗
S1.1 32B	simplescaling/s1.1-32B	✓
o1 (2024-12-01)	-	✓

Table 3 All datasets included in AbstentionBench. **Scenario key:** AU = Answer Unknown; FP = False Premise; S = Stale; UC = Underspecified Context; UI = Underspecified Intent. **Format key:** TF = True or false; MC = Multiple-choice; OE = Open-ended.

Name	Scenario	Domain	Format	License
ALCUNA [58]	UC	Biology	TF	MIT
BBQ [59]	UC	Stereotypes	OE	CC-BY-4.0
BIG-Bench (BB)/Disambiguate [60]	UC	General	MC	Apache 2.0
BB/Known Unknowns [60]	AU	General	OE	Apache 2.0
CoCoNot (CCN)/False Presumptions [27]	FP	General	OE	MIT
CCN/Humanizing [27]	S	General	OE	MIT
CCN/Incomprehensible [27]	UI	General	OE	MIT
CCN/Subjective [27]	S	General	OE	MIT
CCN/Temporal [27]	S	General	OE	MIT
CCN/Unknowns [27]	AU	General	OE	MIT
CCN/Unsupported [27]	AU	General	OE	MIT
FalseQA [61]	FP	General	OE	Not specified
FreshQA [62]	S	General	OE	Apache 2.0
GPQA-Abstain (from GPQA-Diamond) [8]	UC	Science	MC	CC-BY-4.0
GSM8K-Abstain (from GSM8K) [7]	UC	Math	OE	MIT
Known Unknown Questions (KUQ)/Ambiguous [16]	UI	General	OE	MIT
KUQ/Controversial [16]	S	General	OE	MIT
KUQ/False Premise [16]	FP	General	OE	MIT
KUQ/Future Unknown [16]	AU	General	OE	MIT
KUQ/Unsolved Problem [16]	AU	General	OE	MIT
MediQ [63]	UC	Medicine	MC	CC-BY-4.0
MMLU-Math-Abstain (from MMLU) [9]	UC	Math	MC	MIT
MoralChoice [64]	S	Philosophy	MC	MIT
MuSiQue [89]	UC	General	OE	CC-BY-4.0
$(QA)^2$ [65]	FP	General	OE	Apache 2.0
QASPER [66]	UC	Computer science	OE	CC-BY-4.0
SituatedQA/Geo [67]	UI	Geography	OE	Not specified
SQuAD 2.0 [68]	UC	General	OE	CC-BY-SA-4.0
Underspecified Math Word Problems (UWMP) [57]	UC	Math	OE	Not specified
WorldSense [69]	UC	General	MC	CC-BY-NC 4.0

D Additional results

D.1 General abstention performance

In Fig. S1 we show abstention precision of frontier LLM models across all `AbstentionBench` datasets. We note that on most datasets the precision is close to 1 for most models—i.e., models rarely over-abstain. In Fig. S2 we show abstention F1 score—which balances recall and precision—and note that the rank ordering of models using F1 mostly agrees with ranking according to recall.

In Fig. S3 we observe that the correlation between response correctness and abstention recall varies substantially across different datasets.

In Table 4 we show average correctness and average abstention recall for each model across all datasets, sorted by decreasing correctness. We note that reasoning models like DeepSeek R1 Distill, s1.1 and o1 are the top 3 performing LLMs. At the same time, DeepSeek R1 Distill and s1.1 are close to the worst models in terms of abstention performance.

Table 4 Average accuracy and average abstention recall for each model.

Model Name	Average Accuracy	Average Abstention Recall
DeepSeek R1 Distill Llama 70B	0.81	0.46
o1	0.80	0.66
S1.1 32B	0.80	0.43
Llama 3.1 70B Tulu 3 DPO	0.79	0.67
Llama 3.1 70B Tulu 3 PPO RLVF	0.79	0.66
Llama 3.3 70B Instruct	0.78	0.66
Gemini 1.5 Pro	0.77	0.67
GPT-4o	0.75	0.69
Qwen2.5 32B	0.75	0.71
Llama 3.1 8B Tulu 3 PPO RLVF	0.75	0.51
Llama 3.1 405B Instruct	0.74	0.68
Llama 3.1 8B Tulu 3 DPO	0.74	0.53
Llama 3.1 70B Instruct	0.74	0.64
Llama 3.1 70B Tulu 3 SFT	0.70	0.57
Llama 3.1 8B Instruct	0.70	0.66
Mistral 7B v0.3	0.69	0.63
Llama 3.1 8B Tulu 3 SFT	0.65	0.43
OLMo 7B	0.56	0.54
Llama 3.1 70B Base	0.50	0.49
Llama 3.1 8B Base	0.42	0.44

D.2 Effect of scale

In Section 4.1 we discuss the limited effect of increasing Llama 3.1 model scale on abstention recall. In Fig. S4 we additionally provide abstention F1 score, abstention precision, and response accuracy, showing a limited effect of scale across all metrics.

D.3 Effect of post-training

In Section 4.2 we see that Tülu [77] post-training generally improves Llama 3.1 8B abstention performance with the exception of samples with underspecified contexts, and that the majority of performance improvements are observed during SFT and DPO, with PPO RLVF degrading abstention performance.

In Fig. S5 we additionally present abstention F1 score and precision, noting degraded precision (i.e., over-abstention) for questions about stale data. Fig. S6 shows consistent results at 70B scale.

We additionally compare Llama 3.1 Instruct models against their underlying base models, where the instruction-tuned models have undergone multiple successive rounds of both SFT and DPO [73]. We find results broadly consistent with our Tülu observations at 70B scale in Fig. S8, with underspecified context samples proving challenging, alongside subjective questions. At 8B scale, Llama instruction tuning generally improves